

OSH GLOBAL SYSTEMS

ARCHITECTURAL RUNTIME BENCHMARK 2

Exact Replay Technical Report

OSH structured homeostatic cycle vs. TinyLlama 1.1B Chat Q4_K_M

CPU-only | 30 matched inputs | one-token Transformer baseline | same benchmark harness

OSH MEDIAN	RUNTIME GAP	MEMORY GAP
28.760 ms	631.45x	26.88x
30 measured cycles	TinyLlama / OSH median	TinyLlama / OSH Working Set

Integrity status The exact replay harness was verified byte-identical to the original architectural runtime benchmark. Harness SHA-256: 8ac89fef79b65cf11822c4492d0fbefe986239b975a155336dd7a878fd015918.

Interpretation boundary This is an architectural runtime and resident-memory benchmark. It does not score intelligence, reasoning quality, factual accuracy, or final language quality.

Report date: 20 August 2026
Hardware: Intel Atom D2500 @ 1.86 GHz | 8,172.4 MB physical RAM
Execution platform: Windows | CPU only

1. Executive Summary

Benchmark 2 was executed as an exact replay of the previously validated OSH vs. TinyLlama architectural runtime benchmark. The purpose was to measure local computational cost under the same harness rather than to introduce a new scoring method.

The OSH side executed a structured homeostatic organ cycle for 30 fixed inputs and was accepted only when the input was correctly received and the required structured state was returned. The Transformer side used TinyLlama 1.1B Chat Q4_K_M through llama.cpp, CPU-only, with generation deliberately limited to exactly one token per input.

OSH MEDIAN 28.761 ms P95 34.726 ms	TINYLLAMA MEDIAN 18,160.694 ms P95 41,119.876 ms	MEDIAN RATIO 631.45x TinyLlama / OSH
OSH LOADED WS 24.668 MB resident memory after load	TINYLLAMA READY WS 662.977 MB server ready resident memory	MEMORY RATIO 26.88x TinyLlama / OSH

The exact replay produced an OSH median of 28.760 ms and a TinyLlama engine median of 18,160.694 ms, yielding a 631.45x median runtime ratio. Resident Working Set was 24.668 MB for OSH versus 662.977 MB for TinyLlama, a 26.88x memory ratio.

Relative to the historical Benchmark 1 reference, the exact-replay median is effectively in the same regime (+1.85%), while OSH P95 improved from 43.519 ms to 34.726 ms (-20.21%). The TinyLlama median changed by only about +0.10%, which supports the view that the machine-level comparison remained broadly stable.

Key conclusion Benchmark 2 primarily demonstrates reproducibility. The central runtime gap remains on the order of hundreds of times under the benchmark definition, while the OSH tail latency is materially lower than in the earlier reference run.

2. Objective and Scope

The benchmark was designed to compare computational runtime cost and resident memory, not semantic quality. This distinction is essential because the two systems execute different computational architectures and the Transformer baseline is intentionally minimized to one generated token.

2.1 What is measured

- OSH cycle latency for `FaroOshLanguageOrgan.procesar_organo_lenguaje(input, None, "")`.
- Presence of structured OSH outputs: `base_state`, `deformed_state`, `state`, `captures`, and `speech_act`.
- C/P/X/T/H fields inside the required state objects.
- OSH resident Working Set after the benchmark module, instance, and method are loaded.
- TinyLlama engine timing, with HTTP wall-clock retained as a secondary measurement.

- TinyLlama process-tree Working Set when the server is ready.

2.2 What is explicitly not measured

- Intelligence or general reasoning capability.
- Factual accuracy.
- Final language quality or human preference.
- Token throughput beyond the single-token Transformer baseline.
- GPU acceleration; both sides are evaluated CPU-only.

Black-box protection This report documents the benchmark interface, validation rules, timing boundaries, and results. It does not disclose proprietary OSH internal mechanisms beyond the public structured-cycle interface required to reproduce the measurement.

3. Test Environment

Parameter	Value
CPU	Intel(R) Atom(TM) CPU D2500 @ 1.86 GHz
Physical RAM	8,172.4 MB
OSH inputs	30 measured + 3 warmups
Transformer inputs	30 measured + 3 warmups
GPU layers	0
TinyLlama quantization	Q4_K_M
Transformer generation	Exactly 1 token per measured input
Harness integrity	Exact replay of original benchmark; byte-identical copy
Harness SHA-256	8ac89fef79b65cf11822c4492d0fbefe986239b975a155336dd7a878fd015918

4. Exact Replay Implementation

Benchmark 2 reused the original benchmark harness without changing the timing logic. This is important because even small instrumentation changes can alter process memory or introduce timing overhead. The exact replay therefore serves as the formal comparison instrument.

4.1 Process isolation

The OSH phase runs in a separate Python child process. TinyLlama runs in a separate llama.cpp server process. The benchmark parent orchestrates the two phases sequentially and writes the final report files only after both phases complete.

4.2 OSH execution path

```
install_audio_stubs()
load_exact("faro_osh_language_organ_benchmark", ORGAN_PATH)
disable_writes(organ)
instance = FaroOshLanguageOrgan()
instance._speak = no_op
method = instance.procesar_organo_lenguaje

# 3 warmups - validated, not timed
for prompt in WARMUP:
    method(prompt, None, "")

# 30 measured inputs
for prompt in PROMPTS:
    t0 = perf_counter()
    value = method(prompt, None, "")
    elapsed_ms = (perf_counter() - t0) * 1000
    validate_structured_cycle(prompt, value)
```

Known persistence functions are replaced in memory with no-op functions for the benchmark module. Audio output is also disabled. The benchmark does not modify FARO/OSH source files.

4.3 OSH validation gate

- Returned input must exactly match the measured prompt.
- `base_state`, `deformed_state`, `state`, `captures`, and `speech_act` must all exist.
- `base_state`, `deformed_state`, and `state` must each contain C, P, X, T, and H.
- At least five distinct state signatures must appear across the 30 measured inputs.

The exact replay passed the structural gate with 9 distinct state signatures across 30 measured inputs.

4.4 Memory instrumentation

Windows `PROCESS_MEMORY_COUNTERS_EX` is read through `OpenProcess` and `GetProcessMemoryInfo`. For OSH, the comparison uses the loaded Working Set measured after loading the module, creating the instance, disabling voice, and resolving the target method, but before the three warmups. Per-input Working Set samples are also collected to track sampled peak memory.

4.5 TinyLlama baseline

TinyLlama 1.1B Chat Q4_K_M is served by llama.cpp on localhost with GPU layers set to zero and context size 1024. Each request uses temperature 0, seed 42, and `max_tokens = 1`. This deliberately minimizes generation work and therefore makes the Transformer baseline conservative rather than favorable to OSH.

5. Measured Results

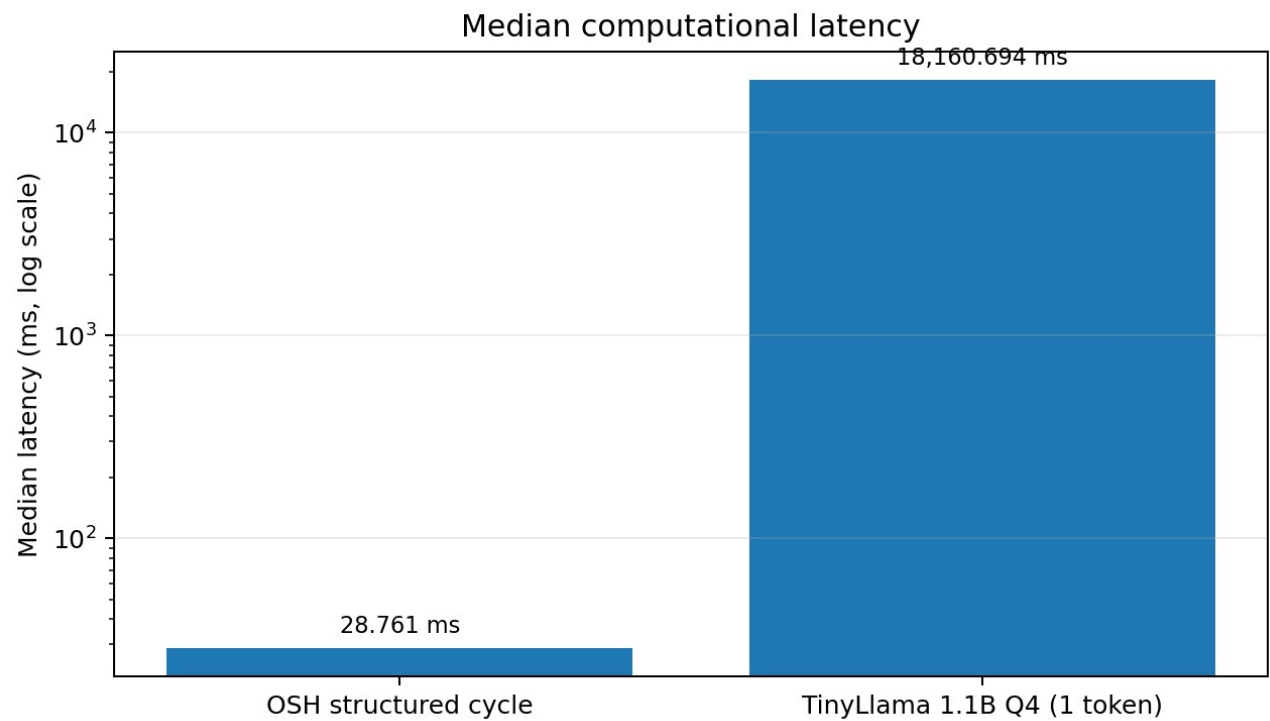


Figure 1. Median latency comparison on a logarithmic axis.

Metric	OSH	TinyLlama	TinyLlama / OSH
Median latency	28.761 ms	18,160.694 ms	631.45x
Mean latency	26.511 ms	21,298.421 ms	803.39x
P95 latency	34.726 ms	41,119.876 ms	1184.14x
P99 latency	40.043 ms	43,361.286 ms	1082.87x
Minimum	17.872 ms	1,518.286 ms	84.95x
Maximum	41.299 ms	43,621.470 ms	1056.24x
Median-derived rate	34.77 cycles/s	0.05506 cycles/s	631.45x OSH rate

5.1 OSH latency profile

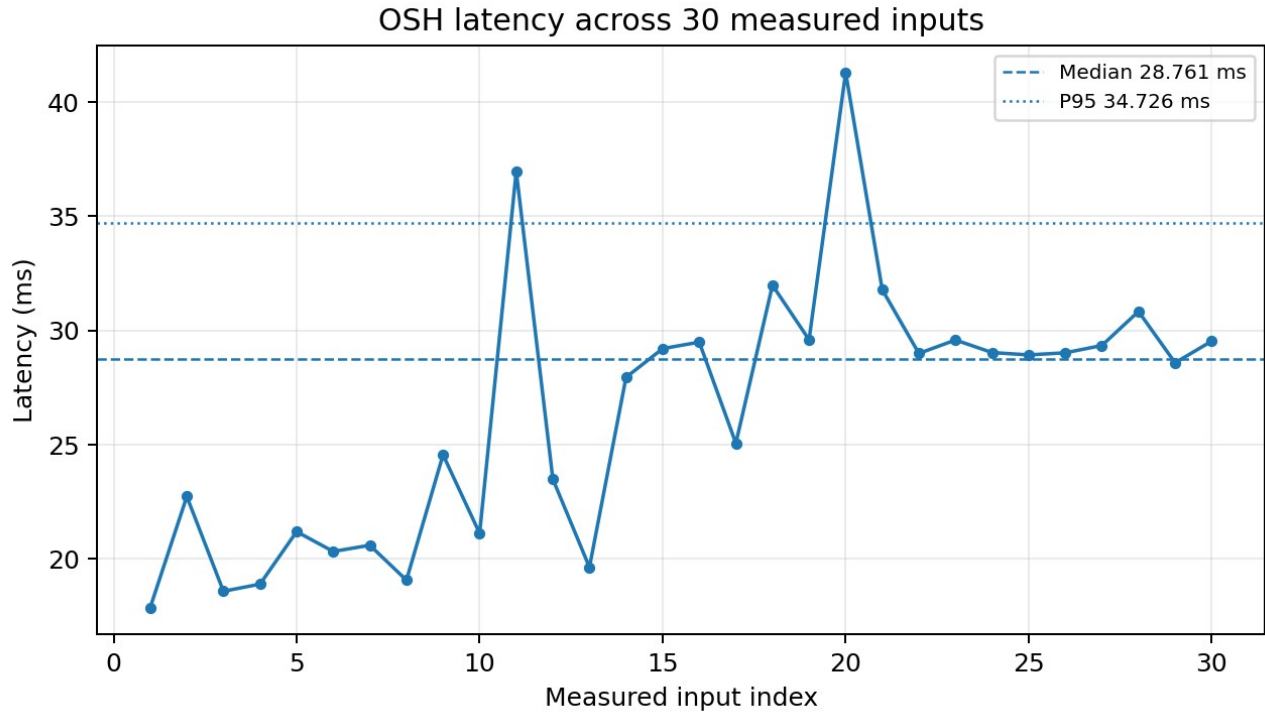


Figure 2. OSH latency for all 30 measured prompts. Dashed and dotted lines show the median and P95.

OSH median latency was 28.761 ms, mean latency 26.511 ms, and P95 34.726 ms. The fastest measured cycle was 17.872 ms and the slowest was 41.299 ms.

Nine distinct homeostatic state signatures were observed. This exceeds the benchmark validity threshold of five and confirms that the run was not accepted as a degenerate constant-state execution.

5.2 Resident memory

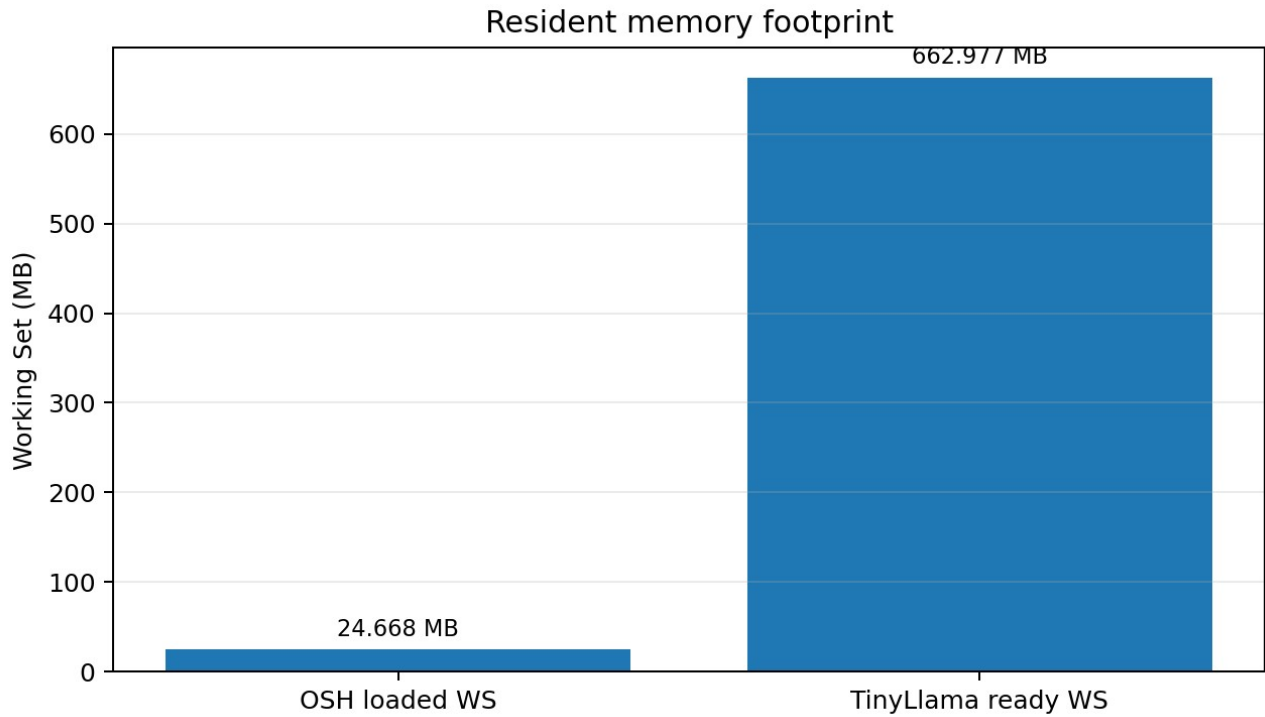


Figure 3. Resident Working Set used for the benchmark comparison.

OSH loaded Working Set was 24.668 MB. TinyLlama ready Working Set was 662.977 MB. The resulting resident-memory ratio was 26.88x in favor of OSH under this benchmark definition.

Important memory definition The OSH figure is the absolute Working Set of the benchmark child process after the OSH benchmark module and instance are loaded. It is not a subtraction-based estimate of the organ alone. Likewise, the TinyLlama figure is the ready process-tree Working Set. This is the same definition used by the original harness.

6. Reproducibility Against Benchmark 1

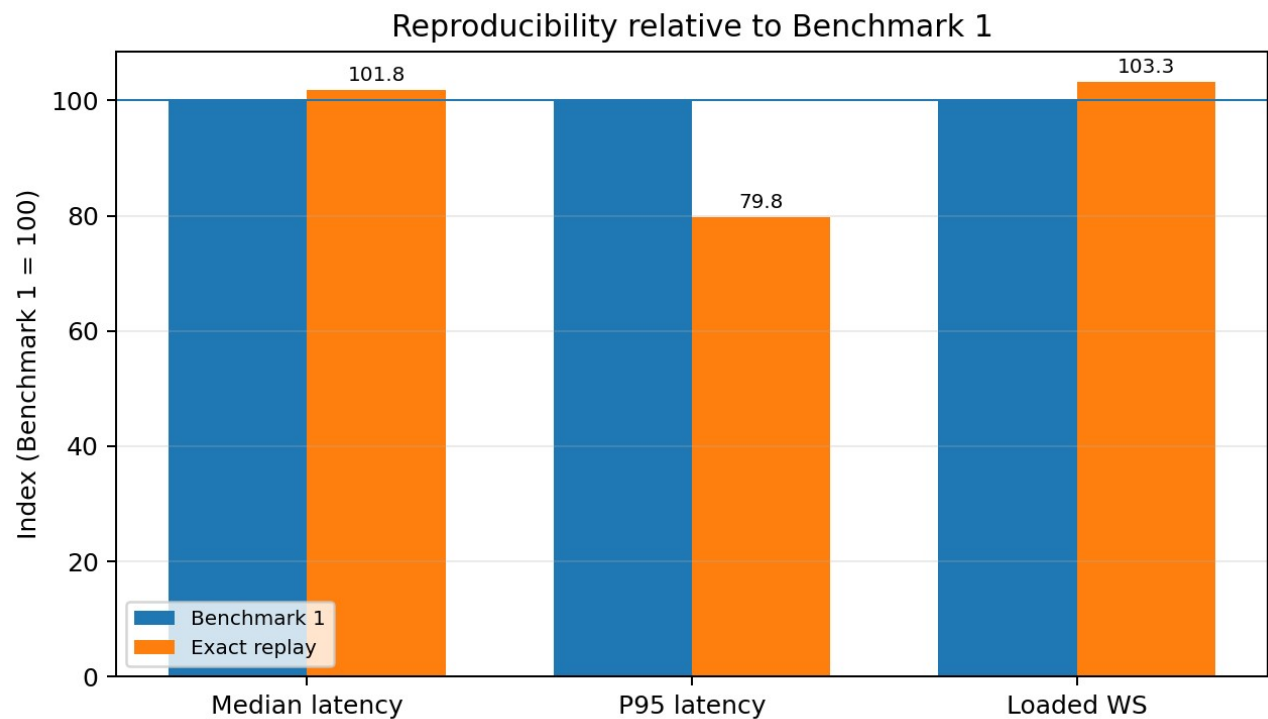


Figure 4. Exact replay normalized to the earlier Benchmark 1 reference (Benchmark 1 = 100).

Metric	Benchmark 1	Exact Replay	Change	Interpretation
OSH median	28.239 ms	28.760 ms	+1.85%	Same operating regime; no median improvement demonstrated.
OSH P95	43.519 ms	34.726 ms	-20.21%	Materially lower tail latency.
OSH loaded WS	23.883 MB	24.668 MB	+3.29%	Small absolute process-memory variation.
TinyLlama median	18,141.751 ms	18,160.694 ms	+0.10%	Baseline remained highly stable.
TinyLlama ready WS	663.930 MB	662.977 MB	-0.14%	Baseline memory remained highly stable.
Runtime ratio	642.43x	631.45x	-1.71%	Hundreds-fold runtime gap

Metric	Benchmark 1	Exact Replay	Change	Interpretation
				reproduced.
Memory ratio	27.80x	26.88x	-3.31%	~27x memory gap reproduced.

The near-zero change in the TinyLlama median (+0.10%) is useful context: it suggests that the machine and baseline runtime conditions were broadly comparable between the historical reference and the exact replay. OSH median latency likewise remained in the same regime, while the P95 result improved substantially.

The exact replay therefore supports reproducibility more strongly than it supports a claim of lower median latency. Any claim that the current OSH architecture has become faster should be tested with a benchmark designed to isolate the evolved runtime path under a fixed methodology, rather than inferred from the UI telemetry alone.

7. Technical Interpretation

7.1 Runtime gap

At the median, TinyLlama required approximately 18.161 seconds of engine time for prompt ingestion plus one generated token, while OSH completed its structured cycle in approximately 28.760 milliseconds. The measured ratio is 631.45x.

Because TinyLlama is restricted to one output token, the comparison intentionally avoids the large additional cost of generating a normal multi-token answer. The result should therefore be read as a conservative local runtime-cost comparison, not as a comparison of equivalent language output.

7.2 Memory gap

The resident-memory result shows a 24.668 MB loaded Working Set for OSH and 662.977 MB for the TinyLlama server process tree, producing a 26.88x ratio. This captures process-resident memory under the harness rather than model file size or disk storage.

7.3 Stability and tail behavior

The OSH exact-replay P95 of 34.726 ms is notably below the historical 43.519 ms reference. Although a single pair of runs is not sufficient to prove a permanent distributional change, the result indicates that the replay exhibited a tighter high-latency tail.

7.4 Relation to live UI telemetry

The live UI measurements observed immediately before this benchmark were approximately 18-21 ms with an 18.0 MB worker Working Set. Those values are valid for the V5/V6 UI measurement protocol, but they are not interchangeable with the exact-replay benchmark because the UI worker deliberately freezes selected external state reads to prevent cross-turn contamination. This report therefore uses only the exact-replay harness for the formal Benchmark 2 comparison.

8. Limitations and Correct Use of the Result

- The workload contains 30 fixed prompts; it is not a broad application benchmark.

- TinyLlama generates exactly one token; normal conversational generation would take longer, but this benchmark intentionally does not measure that.
- Language quality is not scored for either system.
- The OSH result validates structured state and speech-act production, not the quality of the visible sentence.
- The benchmark uses one low-power CPU platform. Results on other hardware should not be assumed from this run.
- Working Set is a process-resident memory metric and can vary slightly across runs because of allocator, loader, and operating-system behavior.
- The runtime ratio compares two different architectures at a defined computational boundary; it should not be presented as an intelligence ratio.

Recommended public wording On the tested Intel Atom D2500 CPU, the exact-replay benchmark measured a 28.760 ms median OSH structured-cycle latency versus 18,160.694 ms for a TinyLlama 1.1B Q4_K_M one-token CPU baseline, a 631.45x median runtime gap. Resident Working Set measured 24.668 MB versus 662.977 MB, a 26.88x gap. The benchmark does not score intelligence or language quality.

9. Reproduction Procedure

The exact replay can be reproduced on the same environment by executing the byte-identical harness. The benchmark itself writes a timestamped folder under Downloads containing JSON, CSV, plain-text summary, and an HTML investor card.

Command

```
python "%USERPROFILE%\Downloads\OSH_VS_TINYLLAMA_ARCH_RUNTIME_BENCHMARK_2_EXACT_REPLAY.py"
```

Expected phases: (1) OSH structured homeostatic cycle for 30 inputs; (2) TinyLlama conservative CPU baseline for the same 30 inputs; (3) aggregate runtime and Working Set comparison; (4) report file generation. The console must end with "FARO files modified: NONE" for a clean run.

Appendix A - Fixed Input Set

#	Prompt
1	hola
2	como estas?
3	que es la vida?
4	que es el sistema solar?
5	por que los planetas no se alejan del sol?
6	que es la fotosintesis?
7	que es la gravedad?

#	Prompt
8	que es una celula?
9	que es la homeostasis?
10	que funcion cumple el corazon?
11	que es una galaxia?
12	por que existe el dia y la noche?
13	que es la energia?
14	que es una orbita?
15	que es el ADN?
16	que significa equilibrio?
17	si una caja tiene 3 bolas rojas y 2 azules cuantas hay?
18	si todos los mamiferos respiran y un perro es mamifero que se concluye?
19	si tengo 10 objetos y retiro 4 cuantos quedan?
20	que diferencia hay entre masa y peso?
21	que es una neurona?
22	que es la memoria?
23	que significa aprender?
24	como funciona un termometro?
25	que es una molecula?
26	por que el agua hierve?
27	que es el tiempo?
28	que diferencia hay entre causa y efecto?
29	que es coherencia?

#	Prompt
30	que es un sistema dinamico?

Appendix B - Raw OSH Measurements

#	Prompt	Latency (ms)	Speech act
1	hola	17.872	presence
2	como estas?	22.734	state
3	que es la vida?	18.583	definition
4	que es el sistema solar?	18.897	definition
5	por que los planetas no se alejan del sol?	21.194	cause
6	que es la fotosintesis?	20.327	definition
7	que es la gravedad?	20.598	definition
8	que es una celula?	19.070	definition
9	que es la homeostasis?	24.560	definition
10	que funcion cumple el corazon?	21.120	function
11	que es una galaxia?	36.968	definition
12	por que existe el dia y la noche?	23.494	cause
13	que es la energia?	19.636	definition
14	que es una orbita?	27.952	definition
15	que es el ADN?	29.208	definition
16	que significa equilibrio?	29.492	definition
17	si una caja tiene 3 bolas rojas y 2 azules cuantas	25.064	quantity

#	Prompt	Latency (ms)	Speech act
	hay?		
18	si todos los mamiferos respiran y un perro es mamifero que se concluye?	31.985	inference
19	si tengo 10 objetos y retiro 4 cuantos quedan?	29.596	quantity
20	que diferencia hay entre masa y peso?	41.299	open
21	que es una neurona?	31.786	definition
22	que es la memoria?	28.998	definition
23	que significa aprender?	29.586	definition
24	como funciona un termometro?	29.036	mechanism
25	que es una molecula?	28.933	definition
26	por que el agua hierve?	29.025	cause
27	que es el tiempo?	29.345	definition
28	que diferencia hay entre causa y efecto?	30.836	cause
29	que es coherencia?	28.588	definition
30	que es un sistema dinamico?	29.536	definition

Appendix C - Raw TinyLlama Measurements

#	Prompt	Wall (ms)	Engine (ms)
1	hola	1,543.892	1,518.286
2	como estas?	16,445.165	16,420.218
3	que es la vida?	16,828.199	16,792.312

#	Prompt	Wall (ms)	Engine (ms)
4	que es el sistema solar?	18,063.889	18,033.049
5	por que los planetas no se alejan del sol?	43,647.944	43,621.470
6	que es la fotosintesis?	29,668.070	29,639.646
7	que es la gravedad?	18,010.883	17,984.745
8	que es una celula?	19,293.453	19,267.420
9	que es la homeostasis?	18,126.546	18,099.368
10	que funcion cumple el corazon?	21,691.360	21,665.638
11	que es una galaxia?	17,941.253	17,913.469
12	por que existe el dia y la noche?	22,965.518	22,940.583
13	que es la energia?	16,592.093	16,566.156
14	que es una orbita?	16,383.529	16,358.445
15	que es el ADN?	16,584.376	16,559.041
16	que significa equilibrio?	20,674.686	20,650.999
17	si una caja tiene 3 bolas rojas y 2 azules cuantas hay?	39,184.325	39,158.932
18	si todos los mamiferos respiran y un perro es mamifero que se concluye?	42,749.262	42,724.285
19	si tengo 10 objetos y retiro 4 cuantos quedan?	36,699.054	36,673.686
20	que diferencia hay entre masa y peso?	24,818.149	24,790.935

#	Prompt	Wall (ms)	Engine (ms)
21	que es una neurona?	16,695.594	16,666.160
22	que es la memoria?	18,246.845	18,222.019
23	que significa aprender?	16,662.408	16,637.250
24	como funciona un termometro?	21,445.597	21,419.882
25	que es una molecula?	17,768.941	17,739.782
26	por que el agua hierve?	19,065.207	19,040.507
27	que es el tiempo?	15,152.848	15,128.572
28	que diferencia hay entre causa y efecto?	19,267.725	19,242.998
29	que es coherencia?	17,463.107	17,438.553
30	que es un sistema dinamico?	20,061.135	20,038.220

Appendix D - Evidence and Integrity Statement

Primary evidence for this report consists of the operator-provided exact-replay console output and the original benchmark harness used to construct the byte-identical replay file. The replay harness SHA-256 was verified as 8ac89fef79b65cf11822c4492d0fbefe986239b975a155336dd7a878fd015918.

The benchmark reported "FARO files modified: NONE." No claim in this document requires disclosure of OSH proprietary internal implementation beyond the measured interface and validation contract.

Exact replay output directory reported by the run: %USERPROFILE%\Downloads\OSH_ARCH_RUNTIME_BENCHMARK_20260820_224056